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AI Translation and Machine Translationese: Argumentation Patterning in CJEU Judgements

1 Introduction

The present paper investigates the use of AI-based technology in the translation of legal discourse with particular emphasis on the translation of argumentative metadiscursive devices in judgements of the European Court of Justice (CJEU). The study focuses specifically on the behaviour of a baseline MarianMT model and a fine-tuned MarianMT model in the translation of discourse-organizing expressions that structure argumentation and guide textual interaction in institutional legal writing. These metadiscursive devices play a central role in the organisation of judicial reasoning because they contribute to inferential sequencing, procedural framing and the presentation of institutional authority in legal discourse.

The principal research question addressed in this paper is whether AI-based neural machine translation systems produce recurrent discourse features signalling argumentation development that differ systematically from human translations and that may be attributable to the algorithmic architecture and training conditions of neural MT systems. To describe such tendencies, the paper draws on the notion of machine translationese, understood as recurrent linguistic characteristics associated specifically with machine-translated output. The concept develops from earlier research on translation universals and translationese in corpus-based translation studies [1, 2]. Machine translationese has been considered a distinct form of text production [3–5] associated, among other things, with reduced linguistic diversity, the algorithmic amplification of frequent patterns [4] and the differences between human and machine translation that can be observed at the level of discourse coherence [6].

In the present study, this notion is operationalised through the controlled comparison of human and machine translations of the same French source judgements. Since the human

translations, baseline translations and fine-tuned translations are all derived from the same source texts, recurrent properties determined primarily by the French source should be reflected across the three datasets. By contrast, recurrent distributional differences shared by the two machine-translated datasets but not found, or found to the same extent, in the human translations provide evidence for machine translationese within this corpus. The comparison between the baseline and fine-tuned systems provides a further distinction: patterns shared by both systems can be separated from changes that emerge or become stronger or weaker following domain-specific fine-tuning. The latter are treated as effects associated with domain adaptation rather than as general properties of machine translationese.

To explore the potential discourse-level manifestations of machine translationese empirically, a corpus of CJEU judgements was translated from French into English using both a baseline MarianMT [7] system and a fine-tuned MarianMT model trained on domain-specific legal material. Methodologically, the study combines high-performance computing (HPC) infrastructure for model fine-tuning and large-scale translation with the corpus analysis of distributional features of metadiscursive devices. The focus on CJEU judgements is motivated by earlier research demonstrating the highly conventionalised and rhetorically structured nature of CJEU legal discourse, particularly in relation to metadiscursive and argumentative organisation [8, 9]. The present analysis concentrates on the French–English language pair because the sufficient data were available for model fine-tuning and systematic comparison. The inclusion of additional language pairs was not feasible within the current stage of the project due to data limitations and computational constraints.

The results presented in this paper should be regarded as exploratory. The study aims to provide an initial framework for identifying discourse-level manifestations of machine translationese in AI-based legal translation and to contribute to the emerging discussion of how neural MT systems reproduce, adapt, or restructure institutional discourse conventions.

2 Linguistic features of EU case law

The Directorate-General for Multilingualism of the Court of Justice of the European Union (CJEU) is responsible for the translation and interpreting activities necessary for the functioning of the Court’s multilingual judicial system [10]. Translation within the CJEU is carried out primarily by lawyer-linguists who are qualified legal professionals with advanced multilingual competences

and specialised expertise in legal translation [11]. As Šarčević [12] argues, legal translation requires not only bilingual competence but also comparative legal knowledge and an understanding of legal reasoning and legal effects. The legal expertise of lawyer linguists is, therefore, central to the translation process because the task involves not merely linguistic transfer but also the preservation of legal equivalence, doctrinal consistency and institutional coherence across multiple legal systems and linguistic traditions.

Like other EU institutions the Court provides specialised linguistic and legal training for translators and lawyer-linguists working within the judicial system. In practice, lawyer-linguists frequently work across several languages, and the multilingual environment of the Court requires continuous interaction between legal expertise and linguistic mediation. As a general institutional principle, translators work into their mother tongue or dominant language [13] and this principle is intended to maximise stylistic naturalness, terminological precision and legal clarity in the target text. CJEU judgements exist in all twenty-four official languages of the European Union. However, these texts are not produced through a simple source–target translation process. Rather, they emerge through a highly complex institutional workflow involving multilingual drafting, translation, revision, legal verification and terminological coordination. As McAuliffe [9, 11] demonstrates, CJEU judgements are the result of a multilayered process of multilingual legal production in which legal reasoning and linguistic form are deeply intertwined. The multilingual production process itself might therefore have linguistic impact and leave discursive traces in the final texts.

The production of CJEU judgements involves several stages, each of which may contribute to the emergence of hybrid or institutionally stabilised discourse forms. During judicial proceedings, parties may submit observations and documentary material in any official EU language. These materials are subsequently translated into the internal language of deliberation, which at the CJEU has traditionally been French. As a result, argumentative structures, terminological choices and rhetorical conventions originating in different legal and linguistic traditions may indirectly shape the French draft of the judgement. The final judgement is then translated from French into the remaining official languages. French occupies a particularly important position within the institutional discourse architecture of the Court which is why it was selected as a source language in the present study. In some cases, translation is carried out through relay or pivot languages. The Court operates with a system of pivot languages, including French,

English, German, Spanish and Italian which function as intermediary nodes for language combinations where direct translation capacity may be limited. Research on multilingualism in EU institutions has repeatedly emphasised the importance of relay translation in maintaining the efficiency and coherence of multilingual judicial communication [14]. At the same time, relay translation may contribute to the diffusion and stabilisation of institutional phraseology and discourse structures across languages.

Another important characteristic of CJEU judgements is their strongly intertextual and formulaic nature. Judicial decisions are rarely drafted entirely from scratch. Instead, they frequently incorporate formulations, argumentative patterns and terminological structures derived from earlier case law and institutional drafting conventions. EU legal discourse is therefore characterised by extensive textual reuse, phraseological repetition and discursive conventionalisation. Such repetition contributes to legal consistency, predictability and institutional continuity, but it also reinforces recurrent rhetorical and argumentative templates across judgements. Previous research has demonstrated that EU legal discourse exhibits particularly strong tendencies toward terminological standardisation and formulaicity [15].

The extent to which computer-assisted translation technologies are currently integrated into the production of CJEU judgements remains insufficiently explored in the scholarly literature. Nevertheless, it is well established that EU institutions maintain extensive technological infrastructures supporting multilingual drafting and translation. These infrastructures include translation memories, terminology databases such as IATE (Interactive Terminology for Europe) and various internal linguistic resources designed to ensure terminological consistency across languages and legal domains. In recent years, EU institutions have also increasingly invested in generative AI and neural language technologies [16].

The linguistic characteristics of CJEU discourse must therefore be understood within this broader institutional and technological ecology. CJEU judgements emerge from a multilingual, highly mediated and strongly standardised production environment in which legal reasoning, translation practices, institutional drafting conventions and technological infrastructures interact continuously. These conditions make EU case law particularly suitable for investigating the interaction between human legal translation, institutional discourse conventions and neural machine translation systems.

3 Argumentation structure and discourse organisation

Research on argumentation development has developed along several partially overlapping traditions in philosophy, linguistics, rhetoric, discourse analysis, and, more recently, artificial intelligence and computational linguistics. In philosophy and logic, argumentation has traditionally been studied primarily in terms of inferential structure and rational justification. Classical models conceptualise arguments as relations between premises and conclusions and focus on questions of validity, soundness, justification and defeasibility. [17–20]. In contemporary studies on AI and law, these approaches have been extended through the theory of argumentation schemes, [19, 20], where legal reasoning is represented through generalised inferential templates such as arguments from precedent, expert opinion, practical reasoning, or analogy. Linguistic and discourse-oriented approaches have supported the view that argumentation is not only a formal logic-based relation between propositions but also a dynamic communicative and rhetorical process realised through language. Within linguistic rhetoric analysis [23], discourse analysis [24–26] and systemic functional linguistics [25, 26] researchers have examined using an interactive approach how argumentative relations are linguistically encoded and how texts through linguistic signals guide readers through complex processes of interpretation and evaluation. Argumentation in language is here studied not as abstract inference but within the more general notion of discourse organisation.

From this perspective, argumentative structure is signalled through a broad range of linguistic devices operating at multiple levels of discourse organisation. These linguistic devices are classified according to their function and the kind of relations they signal (e.g. causal, concessive, contrastive, justificatory relations). Particularly relevant for the present study are discourse-interactional approaches which conceptualise argumentation as a dynamically unfolding network of textual relations rather than a static sequence of premises and conclusions. Thus, Hoey [22, 23] argues that argumentative discourse is organised around culturally recognisable textual patterns such as Problem–Solution, Question–Answer and Claim–Response sequences. These patterns function as interactive structures guiding readers through processes of reasoning, evaluation, anticipation, correction and alignment. Unlike abstract argumentation schemes, these models emphasise the rhetorical realisation of argumentative movement in actual discourse. Argumentative discourse rarely consists of isolated inferential units and instead, arguments are typically embedded within larger discourse structures involving elaboration, justification,

exemplification, correction, reformulation, evaluation and counterclaiming. As Belmonte [29] remarks writers construct argumentation through “complex networks of intertwined textual patterns” rather than through isolated premise–conclusion structures. Thus, Claims may be affirmed, denied, corrected, elaborated, justified, or countered in recursively embedded discourse configurations. Furthermore, argumentative sequences themselves may function rhetorically as elaborations, justifications, or exemplifications within larger argumentative structures.

Such discourse-interactional approaches are especially relevant for legal rhetoric and legal interpretation because legal reasoning is inherently dialogical, evaluative and rhetorically organised [30–32]. Judicial opinions, legal pleadings and statutory interpretation typically involve anticipatory rebuttals, embedded counterarguments, evaluative positioning, hypothetical reasoning and recursive justificatory structures extending across large textual units. Purely formal representations of argumentation may therefore fail to capture the linguistic and rhetorical mechanisms through which legal interpretation is actually constructed and negotiated in discourse.

Since discourse-interactional approaches provide a fine-grained framework for exploring how argumentative reasoning is linguistically encoded and dynamically organised in legal discourse the present study will adopt this approach. This framework makes it possible to investigate not only abstract argumentative relations but also the linguistic realisation, rhetorical organisation and dynamic unfolding of legal argumentation. Such approaches have already been applied to the study of legal discourse [33–36]. In particular, the present study builds on the methodology of identifying argumentative patterns in legal discourse in terms of metadiscursive devices used in my previous research on CJEU judgements. This framework will be explained in some details in the following section. Importantly, the analysis of metadiscursive devices does not capture all dimensions of legal argumentation. The present analysis therefore does not claim to capture legal argumentation in its entirety, which also involves dimensions such as intertextual relations, precedent citation, agency attribution and more complex inferential structures.

4 Metadiscourse, textual colligation and formulaicity in the EU case law

Metadiscourse refers to linguistic features through which writers organise discourse, guide readers through the text and express attitudes toward propositional content [24, 35]. Broadly speaking, metadiscursive devices perform two interconnected functions: they structure textual organisation and they position the speaker or writer in relation to the unfolding argument. Metadiscourse has

been extensively studied across academic, institutional and legal genres because it plays a central role in managing coherence, interaction and rhetorical organisation [24, 36, 37]. In legal discourse, metadiscursive structures are particularly important because judicial reasoning depends heavily on the explicit organisation of argumentative progression, evidential framing and inferential sequencing.

As was indicated above, the present study builds on my earlier research examining discourse patterning in CJEU judgements. In these studies, paragraph-initial and sentence-initial expressions emerged as particularly significant indicators of argumentative structure. Corpus analyses demonstrated that these recurrent expressions could be grouped into relatively stable metadiscursive categories. One of the most prominent classes consisted of transition markers associated with consideration–conclusion relations, including expressions such as *it therefore follows from the foregoing*, *it follows from the above* and similar inferential formulations. These expressions perform discourse-organising functions because they signal transitions from the consideration of arguments presented by the parties toward the formulation of the Court’s own legal conclusions. Their importance is heightened by the fact that CJEU judgements generally lack explicit internal headings. As a result, readers rely heavily on recurrent metadiscursive markers to orient themselves within the argumentative progression of the judgement. Such expressions therefore function not merely as stylistic features but as structural indicators of procedural and inferential development within institutional legal discourse.

The strongly formulaic character of EU legal discourse is particularly relevant in the context of machine translation research. Previous work has repeatedly shown that EU legal language exhibits extensive phraseological repetition, conventionalisation and textual reuse. Judicial reasoning in CJEU judgements relies heavily on recurrent argumentative templates, which is realised through institutional phraseology that contribute to the consistency and predictability of legal interpretation.

Finally, the research demonstrated that multiword metadiscursive items in CJEU judgements tend to occur at the beginning of paragraphs and sentences. The preference of lexical items to occur or avoid occurring in specific textual positions was introduced by Hoey [40] and His notion of textual colligation analogous to the notion of collocations is adopted here. In particular, paragraph-initial metadiscursive expressions are not randomly distributed in CJEU judgements. Instead, they cluster around specific argumentative functions. Frame markers were

used to introduce questions, organise argumentative sections and signal answers, while Importance markers highlighted legally significant points in the reasoning process. Evidentials linked the current judgement to prior case law and external legal authorities. These patterns indicate that discourse organisation and legal argumentation are deeply interconnected. The most frequent category included devices that signal a transition in the text between sections evaluating various considerations before conclusions are drawn. Another important aspect of the findings concerns institutional routinisation since the studies showed that the use of metadiscursive signalling devices is not only a stable category but that has been increasingly used over time in CJEU judgements, suggesting a progressive stabilisation of institutional discourse practices. This led to the conclusion that CJEU judgements are produced within a relatively stable argumentative matrix characterised by repetitive discourse-organisational structures.

In this approach, the study of argumentation patterning involves creating a taxonomy of metadiscursive categories. The degree of depth of such taxonomies can be greater or lesser depending on the objectives of the study. Applying a highly detailed taxonomy in the present context would be extremely labour-intensive because the aim is to compare the occurrence of these expressions across three relatively large datasets, all of which contain substantial numbers of examples that require manual coding and interpretation. Furthermore, a highly differentiated taxonomy risks producing categories with unstable or inconsistent boundaries across datasets and translation systems. Such taxonomies also increase the likelihood that distinctions reflect the researcher's own classification decisions rather than robust discourse tendencies present in the data itself. For all these reasons, the present study adopts a deliberately shallow taxonomy consisting of a limited number of broader categories that capture expressions that signal the development of argumentation in one way or another. In other words, the present approach prioritises analytical stability, comparability and the identification of broader distributional patterns over fine-grained functional differentiation.

Category	Core function
Conclusion	inferential progression, argumentative consequence, legal deduction
Contextual_consideration	contextual framing, situating legal reasoning, procedural conditioning

Justification	evidential/procedural support, legal grounding, argumentative legitimisation
Qualification	delimitation, hedging, conditional restriction, institutional caution
Evaluation	assessment, legal evaluation, rhetorical stance toward propositions

Table 1 Categories of metadiscursive devices used in the present study.

These categories capture different functional dimensions through which CJEU judgements organise and develop legal argumentation. *Conclusion* expressions signal inferential progression and the movement toward legal deduction or judicial outcome marking that preceding considerations are being transformed into an interpretative result. *Contextual consideration* expressions provide contextual framing by situating legal reasoning within procedural circumstances, factual background, institutional conditions or previously established argumentative material. *Justification* expressions function as mechanisms of evidential and procedural support, grounding legal claims through references to authority, prior reasoning, treaty provisions, or argumentative legitimisation. *Qualification* expressions introduce delimitation, restriction, hedging, or conditional specification, thereby modulating the scope and applicability of propositions and reflecting the institutional caution characteristic of judicial discourse. Finally, *evaluation* expressions convey assessment and rhetorical stance toward propositions, arguments, or legal situations, signalling degrees of importance, adequacy, relevance, necessity, or validity within the argumentative process.

The present study investigates how such metadiscursive and argumentative structures are reproduced by a baseline and neural machine translation model. In particular, MarianMT is used in the present study. Although smaller than contemporary large language models, MarianMT was selected for three methodological reasons. First, the model includes multilingual administrative and institutional training data, including EU-related material such as texts originating from the European Central Bank and other EU institutions. Second, MarianMT provides a transparent and computationally manageable environment for domain-specific fine-tuning. Third, the use of a comparatively smaller architecture makes it possible to isolate more clearly the effects of legal-domain adaptation on discourse organisation.

The classification was carried out by a single annotator. Category assignment was based on the argumentative function of each expression, as defined by the taxonomy described above. Where the function of an expression could not be determined reliably from the extracted 5-gram alone, its broader textual context in the judgement was consulted. The relatively broad functional categories used here were intended to avoid distinctions that would require more fine-grained and potentially less stable pragmatic interpretation. This is particularly relevant in the highly conventionalised discourse of CJEU judgements, where many of the expressions examined perform recurrent and relatively stable discourse-organising functions.

Importantly, the classification was applied consistently across the three datasets. The study compares the distributions of expressions assigned to the same categories in the human, baseline and fine-tuned translations; category membership was not determined separately for each dataset. Therefore, although the assignment of an individual expression to a category may be disputed, this does not in itself produce a difference between the three translation conditions. Independent annotation would clearly provide a stronger validation of category boundaries and coding reliability. Such validation was beyond the scope of the present study, and the use of a single annotator should therefore be regarded as a limitation of the analysis.

5 Corpus data and the method of fine-tuning

The empirical basis of this study is a domain-specific corpus of legal texts drawn from EUCLCORP [41]. The dataset consists of 200 judgements of the CJEU in French and English. The corpus was constructed according to several criteria. First, only judgements for which both French and English versions were available were included since the study focuses on the comparison between human and machine-translated realisations of argumentative structures across languages. Second, the judgements were selected so as to cover different periods of CJEU jurisprudence, ranging from early judgements of the Court to judgements published in 2011, which corresponds to the temporal coverage of the EUCLCORP corpus. This diachronic distribution was important because CJEU judgements exhibit certain historical shifts in drafting conventions and rhetorical organisation. An additional selection criterion was motivated by the fine-tuning procedure employed in the study. Since the study investigates the extent to which fine-tuning affects the preservation of argumentative structures in machine translation, particular care was taken to avoid data leakage between the evaluation corpus and the training data used for fine-

tuning. Judgements were therefore selected in such a way that they did not contain entire sentences that appeared in the training corpus used for the fine-tuned model. This was necessary in order to ensure that the evaluation reflects genuine generalisation rather than memorisation of previously encountered sentence pairs.

The data were tagged with XML markers indicating the beginning and end of sentences and paragraphs. This enabled the subsequent extraction and comparison of paragraph-initial and sentence-initial expressions across the datasets. Since MT systems tend automatically to translate or otherwise alter XML tags, the Python translation pipeline was specifically designed to preserve the structural markup and exclude XML tags from the translation process itself. The following example illustrates the structure of the annotated data.

```
<SENT_END>
<PAR_END>
<PAR_START>
<SENT_START>
33 according to the terms of the thirteenth recital of the preamble to regulation
no 33,"the competent authorities must be in a position constantly to follow trade
movements in order to assess market trends and to apply the measures...
<SENT_END><SENT_START>
in fact, respect for fundamental rights forms an integral part of the general
principles of law protected by the court of justice.
<SENT_END>
<SENT_START>
the protection of such rights, whilst inspired by the constitutional traditions
common to the member states, must be ensured within the framework of the structure and
objectives of the community.
<SENT_END>
<SENT_START>
it must therefore be ascertained, in the light of the doubts expressed by the
verwaltungsgericht, whether the system of deposits has infringed rights of a fundamental
nature, respect for which must be ensured in the community legal system.
<PAR_START>
<SENT_START>
the first question (legality of the system of deposits)
<SENT_END>
<PAR_END>
<PAR_START>
<SENT_START>
by the first question the Verwaltungsgericht asks whether the undertaking to
export based on the third subparagraph of article 33 (33) of regulation no 33, the
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lodging of a deposit which accompanies that undertaking and forfeiture of the deposit should exportation not occur during the period of validity of the export licence comply with the law.

<SENT_END>

<PAR_END>

<PAR_START>

<SENT_START>

33 according to the terms of the thirteenth recital of the preamble to regulation no 33, "the competent authorities must be in a position constantly to follow trade movements in order to assess market trends and to apply the measures...

<SENT_END>

CJEU judgements come in different formats depending on the year of publication. Consequently, some of the older documents contain OCR-derived representations characterised by inconsistent spacing, fragmented tokens and extensive capitalisation. In order to prevent such features from introducing unnecessary noise into the translation process and subsequent corpus analysis, all texts were normalised prior to processing. The resulting corpus was subsequently cleaned and normalised prior to analysis. The preprocessing stage included normalisation of casing, removal of OCR-related formatting inconsistencies, normalisation of whitespace and punctuation spacing and reconstruction of paragraph boundaries.

In addition, all numerical expressions were replaced with the randomly selected placeholder number "33." This procedure was implemented in order to minimise the influence of highly variable numerical material on both the translation models and the extraction of recurrent discourse-organisational patterns. Legal judgements contain large quantities of article numbers, dates, case identifiers, percentages, monetary amounts and procedural references which are not themselves relevant to the analysis of argumentative and metadiscursive structures. Such numerical variation can artificially inflate lexical diversity, fragment otherwise identical discourse patterns into multiple surface variants and encourage machine translation systems to memorise or reproduce highly repetitive procedural sequences. By replacing all numbers with a uniform placeholder, structurally equivalent expressions could be grouped more reliably, enabling the analysis to focus on the underlying discourse functions rather than superficial numerical variation.

Two experimental conditions were implemented. In the first condition, the selected documents were translated using the baseline MarianMT model without any domain adaptation. In the second condition, the model was fine-tuned on a domain-specific subset consisting of approximately 200,000 French-English aligned sentences extracted from CJEU judgements. Prior

to fine-tuning, the dataset underwent additional preprocessing procedures in order to reduce noise and repetitive structures that could bias the model toward excessive formulaic repetition. In particular, duplicate and near-duplicate sentence pairs were removed because highly repetitive legal formulations may encourage MT systems to overproduce recurrent expressions and generate loop-like discourse patterns in translation output. This preprocessing step was therefore intended to improve the diversity and stability of the generated discourse structures while preserving the characteristic argumentative and institutional style of CJEU judgements.

The outputs of both conditions were then compared to the corresponding human-produced texts. These human texts are treated as reference translations, although it cannot be established with certainty to what extent they may have been assisted by machine translation systems. In the absence of evidence to the contrary, they are considered here as representative of human institutional production.

Model fine-tuning and corpus translation were carried out on the LEO5 high-performance computing infrastructure using an NVIDIA A100 GPU with 40 GB of memory. The model was fine-tuned for three epochs, corresponding to approximately 17,000–18,000 training steps, with a final validation loss of approximately 0.14. Training required approximately 18 minutes. The use of HPC infrastructure made it possible to combine domain-specific model adaptation with large-scale translation and subsequent corpus-based comparison within a computationally practicable workflow.

As was indicated above, previous corpus-linguistic studies of CJEU discourse [9, 42] have shown that argumentative and metadiscursive structures in these judgements are highly formulaic and exhibit strong distributional regularities over longer periods of time. The corpus therefore provides a sufficiently robust empirical basis for investigating the preservation and transformation of argumentative structures in machine translation. Furthermore, the size of the corpus represents a practical compromise between analytical depth and computational feasibility. Processing substantially larger datasets containing thousands of judgements would considerably increase computational costs and processing time even on high-performance computing (HPC) systems, particularly when multiple experimental configurations and iterative analyses are required. Besides, the annotation of metadiscursive categories was carried out manually and for thousands of examples this would be beyond the scope of the present project. The present corpus therefore

enables detailed qualitative and quantitative analysis while remaining computationally manageable.

6 Distributional properties of metadiscursive devices in translated CJEU judgements

After the data has been fine-tuned and translated using LEO5 computational system a set of corpus linguistic studies was carried out. All the investigations were carried out using Python scripts that have been written specifically for the purposes of the present study. The analysis proceeded in several stages.

The first stage consisted of a general comparison of the translated corpora in terms of their basic distributional characteristics. This included comparisons between the human-produced English and French judgements and the outputs generated by the baseline and fine-tuned machine translation systems. Measures such as token counts, type-token ratios and other descriptive corpus statistics were used in order to establish a general overview of the structural properties of the corpora before proceeding to the analysis of argumentative discourse patterns.

The second stage focused on the extraction of recurrent multiword expressions occurring at sentence-initial and paragraph-initial positions. In particular, fixed five-word sequences (5-grams) were extracted. The decision to focus on 5-grams was motivated by previous research on discourse organisation in CJEU judgements [9, 42] which demonstrated that longer expressions generally do not provide additional information for the identification of metadiscursive functions, whereas 5-grams are sufficiently long to capture the rhetorical and argumentative function of recurrent expressions. In the discourse of CJEU judgements, such expressions frequently function as markers of inferential progression, justification, evaluation, contextual framing and argumentative sequencing.

The extracted expressions were subsequently classified according to the taxonomy presented in Section 4. However, only those categories directly related to argumentative development were included in the present study. The classification process combined automated extraction with manual qualitative coding. In total, 914 recurrent expressions extracted from the English judgements of all three datasets (i.e. human translations, baseline MT and fine-tuned MT) were manually examined and categorised by the author. Since the present study is concerned with recurrent and institutionally stabilised argumentative patterns, only expressions occurring at least

five times within a dataset were included in the analysis. This threshold proved methodologically appropriate because the frequency distributions followed the expected Zipfian distribution (for more details see Section 6.3) characteristic of natural language corpora, where a relatively small number of highly recurrent expressions account for a substantial proportion of discourse organisation.

Each occurrence was subsequently counted, and frequencies were aggregated at several analytical levels: per document, per rhetorical category, per positional type and per translation system (human, baseline MT, fine-tuned MT). This aggregation transformed individual textual occurrences into a structured dataset suitable for quantitative and statistical analysis. Importantly, the analysis distinguishes between raw frequency counts and normalised frequencies. In addition to absolute counts, relative frequencies and token-mass distributions were calculated in order to enable comparisons across corpora of different sizes. The notion of token mass was introduced to examine not merely whether certain argumentative categories occur, but also the extent to which they dominate the overall argumentative organisation of the translated texts. Finally, both descriptive and inferential statistical measures were employed in order to compare the distributional behaviour of argumentative expressions across the three systems. These included frequency distributions, overlap measures, relative token-mass comparisons and correlation-based analyses designed to explore the extent to which machine-translated judgements reproduce the argumentative organisation characteristic of human-produced CJEU discourse.

6.1 Basic distributional properties

First, the general distributional properties of the CJEU judgement across all four data sets including the French data will be described. The analysis concerns the total number of paragraphs, sentences, word tokens and word types. As shown in Table 2, the baseline and fine-tuned translations contain the same number of lines, paragraphs and sentences. This clearly indicates that both kinds of machine translation preserve the basic textual properties of the French judgements and indicates that the current MT model is the sentence-aligned oriented. In contrast, the human English translations display slightly fewer number of sentences, paragraphs implying that human translators restructure the discourse organisation more flexibly. At the lexical level, however, the two MT systems show different tendencies despite preserving the same sentence and paragraph segmentation as the French source texts. The baseline translations are slightly shorter than the

human English translations, whereas the fine-tuned translations contain substantially more word tokens. On average, the baseline MT produces judgements that are around 3% shorter (71 words fewer per judgement), whereas the fine-tuned MT increases text length by approximately 8% (221 additional words per judgement).

dataset	total_w ords	total_parag raphs	total_sent ences	avg_words_p er_file	avg_types_p er_file	avg_ttr_pe r_file
human_FR	787306	12273	19693	4100.55	965.89	0.2942
human_EN	771282	12059	19211	4017.09	812.88	0.2542
baseline_EN	764808	12273	19693	3983.38	827.52	0.2619
fine-tuned EN	855798	12273	19693	4457.28	687.68	0.2053

Table 2 General textual properties of CJEU judgements across the four data sets

Measured in terms of TTR, the human English translations show a small reduction in lexical diversity (0.2542) compared to the French judgements (0.2942). The baseline MT system produces a slightly higher average TTR value (0.2619) than the human translations, despite containing fewer total tokens overall. The baseline system, therefore, preserves somewhat greater lexical variability and produces less repetitive output than the human translations. The lowest TTR value (0.2053) is observed in the fine-tuned system which also has the highest total number of tokens. This indicates that the fine-tuned model relies more heavily on repetitive lexical and phraseological patterns, producing more standardised and formulaic output. The result therefore shows greater formulaicity in the fine-tuned output, although the present comparison does not establish whether this is an effect of fine-tuning itself or of the characteristics of the domain-specific data on which the model was fine-tuned.

Tables 2 and 3 summarise the distribution of initial multiword expressions and discourse markers in sentences and paragraphs across English datasets. French judgements are excluded from these observations since fixed multiword expressions do not necessarily correspond across languages, as demonstrated in [9].

Dataset	par_5gr:unique	par_5gr:frequency	par_5gr:ttr	sent_5gr:unique	sent_5gr:frequency	sent_5gr:ttr
human EN	963	3703	0.2601	1332	4975	0.2677
baseline EN	993	4289	0.2315	1405	5964	0.2356
fine-tuned EN	992	4602	0.2156	1441	6430	0.2241

Table 3 Distribution of initial multiword metadiscursive expressions.

dataset	all_5gr:unique	all_5gr:frequency	all_5gr:ttr	disc_markers:unique	disc_mark:frequency	disc_mark:ttr
human EN	2295	8678	0.2645	47	7099	0.0066
baseline EN	2398	10253	0.2339	39	5753	0.0068
fine-tuned EN	2433	11032	0.2205	38	5063	0.0075

Table 4 Distribution of initial multiword metadiscursive expressions.

As can be seen in the tables, both MT systems produce a greater variety of the fixed multiword devices occurring in the sentence and paragraph initial positions than human translations. The increase is more marked for the fine-tuned MT system which exhibits the highest overall frequency values across nearly all categories. While both MT systems generate more distinct paragraph- and sentence-initial expressions, their type-token ratios are consistently lower than in the human translations. This applies especially to fine-tuned systems where the increase in total frequency substantially exceeds the increase in the number of unique expressions. It follows that the model repeatedly draws on a relatively stable repertoire of routine expressions. The higher frequencies of sentence- and paragraph-initial expressions combined with lower diversity values suggest that the model relies more heavily on a smaller set of recurrent legal-discursive routines. This observation is consistent with the broader tendencies identified earlier in the analysis, particularly the increase in contextual framing and qualification structures. The baseline translation also produces more

paragraph and sentence initial expressions than the human translations, but its TTR values remain closer to the human data and substantially higher than those of the fine-tuned system.

Interestingly, discourse markers themselves exhibit a somewhat different distributional behaviour. The human translations contain the broadest inventory of discourse markers as well as the highest overall frequency of these items. The MT systems, in contrast, rely on a smaller repertoire of discourse markers. Importantly, the slightly higher TTR values observed for discourse markers in the MT systems should not be interpreted straightforwardly as evidence of greater rhetorical diversity. Since the systems produce fewer overall discourse-marker tokens, relatively small changes in the number of unique items produce proportionally larger effects on the ratio itself. The relevant tendency therefore concerns not the isolated TTR value, but the overall interaction between repertoire size, frequency concentration and rhetorical distribution.

6.2 Distribution of argumentation-signalling items

Not all the sentence and paragraph initial multiword expressions and discourse perform this function and in what follows the focus will be only on metadiscursive expressions that perform argumentative function. For the present analysis, only the items that occur at least five times were selected in order to capture only the strongly repetitive devices. As a result, the total of 914 items across all three English data sets were identified and classified using the current taxonomy.

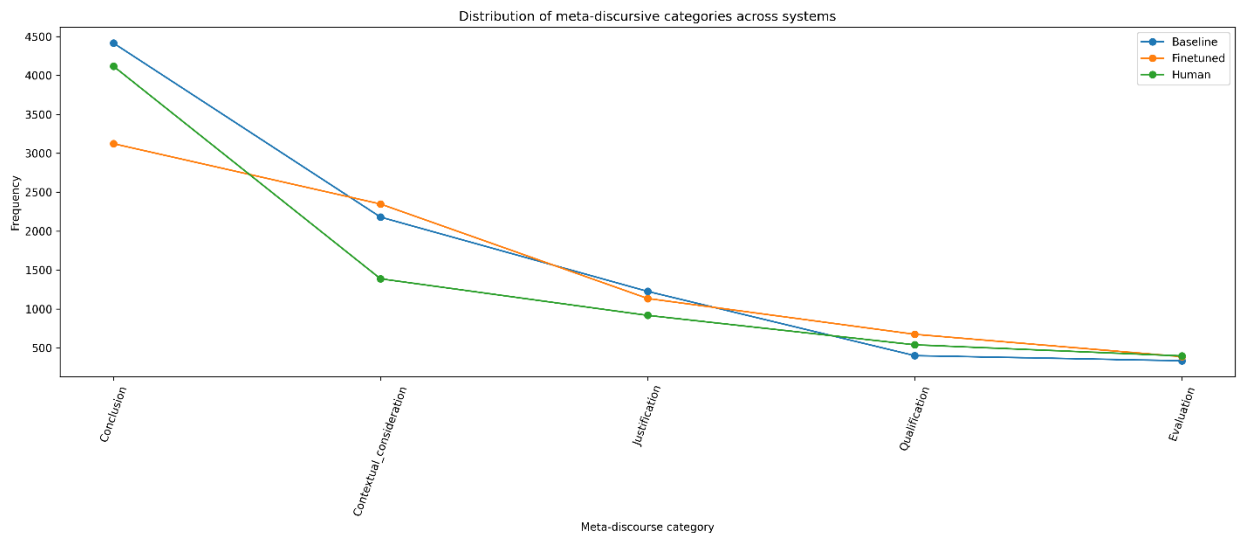


Figure 1 Distribution of metadiscursive expressions across three English datasets. Distribution of metadiscursive expressions across the three English data sets

As can be seen in Figure 1, at a global level, the three datasets exhibit similar tendencies. The Pearson correlation coefficients indicate a strong linear association between the distributions of the machine-translated and human datasets, with the baseline system showing the closer overall correspondence to the human translations ($r = 0.978$) than the fine-tuned system ($r = 0.900$). The correlation between the baseline and fine-tuned systems is similarly high ($r = 0.967$). Moreover, the Spearman correlations are effectively maximal across all comparisons, indicating that the relative ranking of argumentative categories remains almost identical in the three datasets. In other words, all systems broadly agree regarding which metadiscursive functions are globally dominant within CJEU judgements in English.

These correlations indicate a high degree of similarity in the distribution of metadiscursive items across the datasets, particularly in their relative frequency and rank ordering. They should not, however, be interpreted as evidence that the MT systems reproduce the rhetorical functions or the development of argumentation in the same way as the human translations. In fact, there are some important functional differences in the relative prominence of individual categories. All three datasets are strongly dominated by *Conclusion* markers which represent by far the largest proportion of argumentative devices overall. In the baseline system, *Conclusion* signalling devices account for approximately 52% of all argumentative items, compared to 48% in the human translations and only 36% in the fine-tuned system. This confirms that inferential progression and argumentative closure constitute the central organisational principle of CJEU reasoning but also indicates that these argumentative devices are distributed differently across the systems. *Conclusion* signalling devices are more prominent in the baseline system than in the human translations, whereas their relative prominence is considerably reduced in the fine-tuned system. *Contextual consideration* relations, by contrast, account for 25% of the items in the baseline system and 27% in the fine-tuned system, compared with 16% in the human translations. The category of Justification remains comparatively stable across the three datasets accounting for 11% of argumentative devices in the human translations, 14% in the baseline system and 13% in the fine-tuned system. Somewhat stronger differences emerge in the category of *Qualification* devices that account for only approximately 5–6% of the argumentative structures in the human and baseline datasets but increase to around 8% in the fine-tuned system. *Evaluation markers* constitute the smallest category in all three datasets and remain comparatively stable, accounting for approximately 4–5% of all argumentative devices.

6.3 Statistical modelling of category distributions

The number of metadiscursive items (both MWEs and discourse markers) varies across data sets and it is worth considering how much each item contributes to how argumentation develops in CJEU judgements. This allows us to explore whether they contribute equally to the argumentation patterning in judgements or whether some carry more weight.

A relatively small set of recurrent institutional templates was identified at both the paragraph-initial and sentence-initial levels in all three data sets. In the paragraph-initial data, highly frequent expressions such as *therefore*, *it is therefore appropriate to*, *it follows from the foregoing* or *that it is therefore appropriate* occupy prominent positions across all datasets, indicating the existence of a shared inferential core. To examine the stability of this core beyond a fixed frequency threshold, overlap and token-mass coverage were calculated incrementally from the top 1% to the top 50% of ranked expressions. At each step, two measures were computed: (i) Jaccard overlap between systems and (ii) the proportion of total token mass accounted for by the shared expressions. The resulting patterns for paragraph-initial and sentence-initial expressions are presented in Figures 2 and 3 respectively. Whereas the left-hand graph measures how many expressions are shared, the right-hand graphs measure how much of the total discourse is covered by the shared expressions which will be referred to as their token-mass.

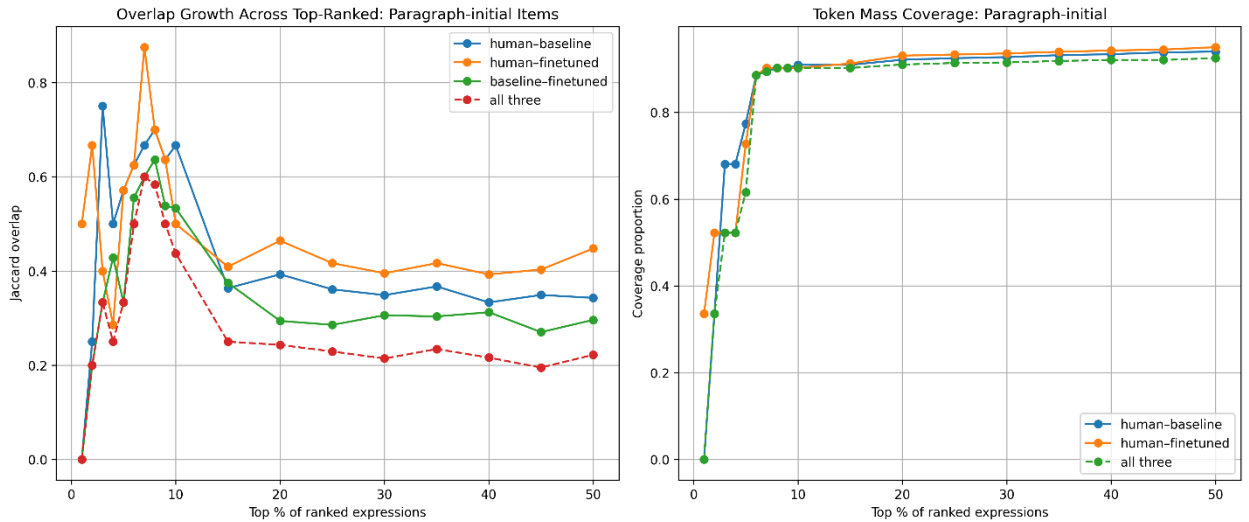


Table 5 Overlap growth of top-ranked paragraph initials and their token-mass.

An important question here is not “How many sentence-initial expressions are shared?” but rather: “How much of the actual running discourse is produced using shared sentence-initial expressions?” The answer is: an extremely large proportion in both sentence and paragraph initial devices. Across both levels of analysis, overlap values fluctuate substantially in the uppermost part of the distributions because the comparison initially involves only a very small number of extremely frequent expressions whose exact lexical composition differs across systems. In the sentence-initial data (Figure 3), this is particularly evident within the top 1–4% of expressions, where overlap values change sharply before becoming progressively more stable. A comparable pattern is found in the paragraph-initial data (Figure 2): although the human–baseline and human–fine-tuned comparisons differ considerably at some of the lowest thresholds, these differences diminish as a larger proportion of the ranked expressions is included. These fluctuations demonstrate that relatively minor differences in the ranking of a handful of dominant expressions can produce large changes in overlap values at the top of the distribution.

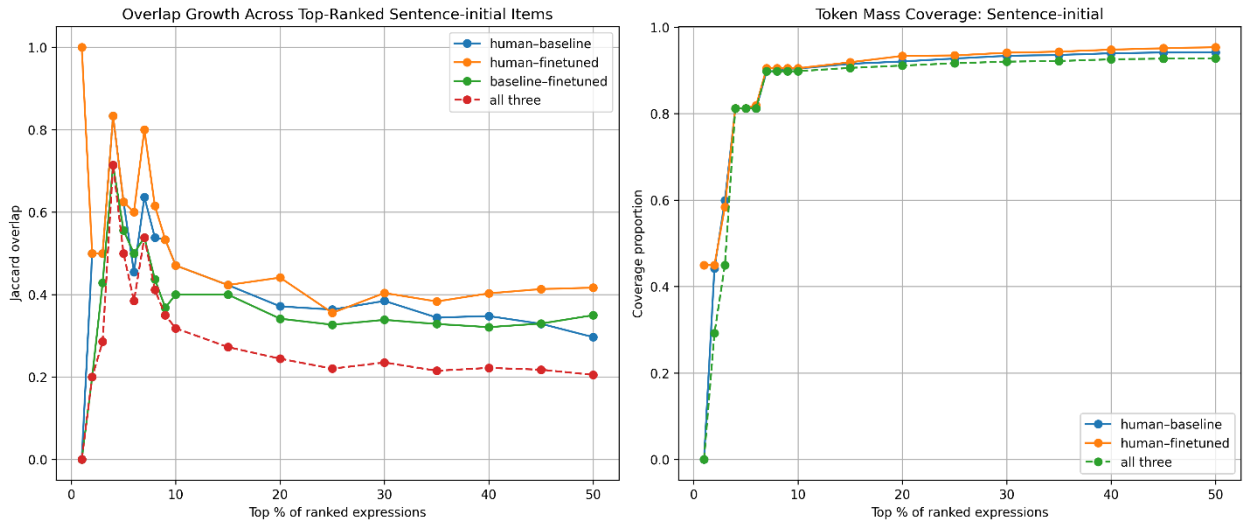


Table 6 Overlap growth of top-ranked sentence initials and their token-mass.

Despite this lexical instability, the token-mass coverage curves reveal a remarkably stable concentration structure across both datasets. Already within the top 5–7% of ranked expressions, the shared items account for approximately 77–90% of the token mass in the human English data. Beyond this point, the coverage curves flatten considerably. By approximately 15–20% of the ranked expressions, the human–fine-tuned comparison covers roughly 91–93% of the token-mass in the human-translated judgements at both the paragraph and sentence levels, while the human–baseline comparison reaches approximately 91–92%. Further increases in the number of included

expressions produce only marginal gains. At 50%, the human–fine-tuned comparison covers approximately 95% of the token mass, whereas the human–baseline comparison reaches approximately 94%. The same overall tendency is therefore visible in both figures: a very small proportion of recurrent expressions accounts for the overwhelming majority of argumentative-metadiscursive usage.

At the same time, overlap values decline steadily beyond the initial high-frequency core. In the paragraph-initial data (Figure 2), overlap stabilises after approximately 15–20% at substantially lower levels, generally around 0.30–0.45 depending on the comparison. The sentence-initial distributions (Figure 3) exhibit an even stronger divergence between overlap and coverage. At 20%, the human–baseline overlap falls to approximately 0.37–0.39, whereas the human–fine-tuned overlap remains somewhat higher at approximately 0.44–0.46. By 50%, the human–baseline overlap declines further to approximately 0.30–0.34, while the human–fine-tuned overlap remains higher at approximately 0.42–0.45. The baseline–fine-tuned overlap stabilises at lower levels throughout the second half of the distributions, generally around 0.27–0.35. The simultaneous increase in token-mass coverage and decrease in lexical overlap indicates that the systems converge strongly in terms of overall argumentative concentration while diverging increasingly in the lexical realisation of lower-frequency expressions.

This divergence between concentration and lexical overlap reflects a strongly Zipfian distributional structure at both levels of analysis. In other words, across all three English data-sets a very limited set of highly frequent argumentative devices carries the overwhelming majority of the load while a comparatively large number of lower-frequency expressions form a variable lexical long tail. In practical terms, this effective core of the system can be identified around the point at which the coverage curves flatten, and additional expressions contribute only minimal increases in token mass. In the present data, this threshold lies approximately between 10% and 15% of the ranked expressions. Beyond this point, differences between systems are driven primarily by low-frequency lexical variation rather than by changes in overall argumentative organisation.

7 Actual translation strategies in human and machine-translated judgements

Despite the general tendencies of convergence observed above, a closer examination of individual examples demonstrates that the same metadiscursive categories are not always realised through

identical expressions. In particular, the data reveal systematic differences between human translation and MT with respect to the interlingual interference. While the human translations frequently restructure the argumentative organisation of the French source text the MT systems tend to preserve the formal inferential architecture of the original text much more directly. This can be observed, for example, in the translation of highly institutionalised inferential operators such as *il résulte de ce qui précède*. Consider the following examples:

1. human_french: *il résulte de ce qui précède qu'en refusant aux succursales et agences en france de sociétés d'assurances ayant leur siège dans un autre état membre...*
2. human_english: *consequently, by failing to grant to the branches and agencies in france of insurance companies whose registered office is in another member state...*
3. baseline_english: *it follows from the foregoing that, by refusing to the branches and agencies in france of insurance companies having their seat in another Member State...*
4. fine-tuned_english: *It follows from the foregoing that, by refusing to grant to branches and agencies in France of insurance companies having their seat in another Member State...*

The human translation restructures the inferential relation from the French expression into a compact adverbial connector (*consequently*), thereby reducing the explicit procedural scaffolding of the original argumentation. By contrast, both MT systems preserve the clause-initial frame. In particular, the baseline system reproduces the French structure almost mechanically, whereas the fine-tuned system additionally modifies grammatical cohesion through the insertion of the missing infinitival complement (*to grant*).

In the examples (5)–(9), the human translation does not reproduce the French impersonal evaluative formula (*il convient de remarquer*) at all. Instead, the argumentative perspective is restructured through contextual embedding into the procedural narrative of the case. The MT systems, on the other hand, reproduce the French impersonal construction directly through the institutional English formula *It should be noted that*. This suggests that MT systems rely heavily on stable bilingual correspondences between recurrent procedural formulae.

5. human_french: *il convient de remarquer qu'il ne suffit pas qu'une réglementation soit assortie de sanctions pénales pour relever de la notion d'ordre public...*

6. human_english: *in this connection the government of the federal republic of germany has submitted first of all that paragraph 33...*
7. baseline_english: *It should be noted that it is not sufficient for a regulation to be accompanied by penal sanctions to fall within the concept of public policy...*
8. fine-tuned_english: *It should be noted that it is not sufficient for a regulation to be accompanied by penal sanctions to fall within the concept of public policy...*

Another representative example concerns the translation of evidential-procedural inferential constructions as illustrated in (9)–(12).

9. human_french: *vu les considérations qui précèdent, il convient de répondre à la juridiction nationale que les articles...*
10. human_english: *It follows from those considerations that the reply to be given to the national court is that Articles...*
11. baseline_english: *Having regard to the foregoing considerations, it must be submitted to the national court that Articles...*
12. fine-tuned_english: *In the light of the foregoing considerations, the answer to be given to the national court must be that Articles...*

The French source text organises the argument through a layered impersonal inferential frame combining evidential reference (*vu les considérations qui précèdent*) with procedural argumentative staging (*il convient de répondre*). The human translation restructures this architecture into a more direct inferential formulation (*It follows from those considerations that*) and suppresses the French impersonal procedural component entirely. The baseline MT output preserves the French structure much more literally through close calques such as *Having regard to the foregoing considerations* and *It must be submitted to the national court*. The fine-tuned translation retains the source-oriented evidential framing but partially restructures the procedural component into the more natural nominal formulation *The answer to be given to the national court must be that*. The fine-tuned system therefore appears to occupy an intermediate position between strict formal preservation and partial rhetorical adaptation.

It is important to emphasise that these examples of translation from French into machine-translated texts are not isolated cases but indicative of a wider trend. This is supported by the fact that some items that do not show up in human translations occur among the top 5% of items in MT systems. One example is the highly recurrent expression *that it is therefore appropriate* that occurs in the baseline output, and which clearly originates from the French procedural constructions such as *qu'il y a donc lieu de répondre que* or *qu'il convient donc de répondre que*. These expressions are extremely frequent in the French judgements and function as institutionalised markers introducing the Court's answer or interpretative conclusion. The baseline system reproduces these structures with a high degree of formal literalness.

13. French: *qu'il y a donc lieu de répondre à la question posée que...*
14. Baseline: *that it is therefore appropriate to answer the question posed that...*
15. Fine-tuned: *The answer to the question referred must therefore be that...*
16. Human: *the question referred should therefore be answered to the effect that...*

The baseline construction is structurally very close to the French source text since it preserves the procedural frame (*il y a lieu de répondre*), the inferential marker (*donc* becomes *therefore*) and even the complementiser-governed structure (*que* becomes *that*).

Similarly, fine-tuned system repeats another recurrent pattern which involves the preservation of concessive argumentative structures introduced by *s'il est vrai que* (see (17)–(20)).

17. French: *s'il est vrai que cette disposition réserve aux États membres...*
18. Human: *although that provision leaves member states to choose...*
19. Baseline: *Although it is true that this provision reserves to the Member States...*
20. Fine-tuned: *Although that provision reserves to the Member States...*

The baseline system reproduces the original French structure very literally through the explicit formula *although it is true that*, preserving the full propositional scaffold of the source expression. The human translation, by contrast, compresses the construction into a simpler concessive clause (*although that provision leaves*), eliminating the semantically weak evaluative component *it is true that*. The fine-tuned system replicates the human translation.

In general, extremely high frequencies of clause-initial subordinate structures occur in the baseline translations, including *that*-initial openings, *whereas*-initial constructions and *since*-initial clauses. The general lexico-grammatical patterns reveal that these are not isolated lexical phenomena but stable transferred discourse templates such as in *that* + NP + VP, *that* + pronoun + VP, *whereas* + clause. These patterns correspond closely to recurrent French legal constructions headed by *que* and related complementiser structures. The baseline system therefore reproduces clause-initial subordination, complementiser-led discourse organisation and French-style inferential chaining within English lexical material. The human translations behave very differently in such cases. Frequencies of clause-initial hypotactic structures are dramatically reduced, particularly in the case of *whereas*-initial constructions, which decline from 217 instances in the baseline output to only 5 in the human translations. Similarly, *that*-initial structures fall from 697 to 177 instances when human and baseline translations are compared. Human translators thus systematically dismantle the French hypotactic architecture rather than reproducing it directly. Instead of preserving subordinate framing structures they reconstruct the discourse according to English legal-rhetorical norms. This restructuring involves several recurrent operations. French hypotactic clauses are frequently converted into nominalised constructions, transforming structures such as *qu'il résulte de* into formulations such as *the result of*. Human translators also compress subordinate clauses into more direct formulations and where MT systems use the structures such as *whereas it is necessary to determine* human translation uses *it is necessary to determine* or nominalised alternatives. Another common strategy involves finite-to-non-finite restructuring, as in the use of *by requiring Member States to* or *Member States being required to*. Instead of *that the Member States are required to*. As a result, these transformations reduce clause dependency, eliminate embedding and shift the discourse toward the more compressed and linear rhetorical organisation characteristic of English institutional prose. The fine-tuned model occupies an intermediate position between direct transfer and target-side adaptation. Fine-tuning substantially reduces the frequency of the most overtly transferred hypotactic structures. *that*-initial openings decline from 697 instances in the baseline system to 281 in the fine-tuned model, while *whereas*-initial constructions fall from 217 to 22. This reduction indicates that the fine-tuned output shifts partially away from French-style clause-initial hypotaxis toward the patterns observed in the human English translations, which rely less heavily on complementiser-led discourse organisation. The present analysis does not, however, establish whether this shift reflects the

acquisition of English institutional conventions more generally or the reproduction of patterns characteristic of the domain-specific fine-tuning corpus.

Another broader pattern that can be observed throughout the dataset is that MT systems tend to preserve the explicit argumentative scaffolding of the French judgements. Inferential operators such as *il résulte de ce qui précède, dans ces conditions* and *des considérations qui précèdent* are translated in a highly stable and repetitive manner as *it follows from the foregoing, in those circumstances* and *from the foregoing considerations*. Human translators, by contrast, frequently compress, omit, or rhetorically restructure these markers through discourse markers such as *therefore, consequently, thus*, or through complete contextual integration into surrounding discourse. In this sense, MT systems replicate both the propositional content and the formalised inferential architecture of the French judgements. Finally, the baseline system frequently produces literal calques such as *must be disparate, must be distinct, or the first purpose of not receiving must be separated*. These examples indicate that French influence persists in either baseline or both MT systems not merely lexically or syntactically but also at the level of argumentative organisation.

8 Discussion and concluding remarks

This paper investigated whether machine-generated translations of judgements of the CJEU exhibit distinctive characteristics. The analysis proceeded from general quantitative distributional patterns toward more detailed examination of recurrent metadiscursive structures. This made it possible to move from broad distributional tendencies to the identification of specific discourse-structural translation strategies operating in the MT and human translations. In particular, building on previous research identifying recurrent metadiscursive strategies in CJEU judgements, the study explored how such structures are preserved, transformed, or reorganised in machine-translated English versions of French judgements. The major results are summarised and discussed below.

At the textual level, the results showed that both MT systems closely copy the main linguistic features of the French source texts. The baseline and fine-tuned outputs had exactly the same number of sentences and paragraphs as the French originals and they depart slightly from the human-translated judgements that tend to be shorter in English than in French. This feature alone indicates that human translators are more flexible in organising discourse, while both MT systems adhere more strictly to sentence and paragraph boundaries. In terms of text length, the baseline system produces texts that are slightly shorter than human translations and concerning

lexical variability, baseline translations exhibit comparable lexical variability to human translations. In contrast, the fine-tuned system generates longer, more repetitive and lexically less diverse texts than the other two English versions. All three types of translation show a lower degree of lexical variation than the French source texts.

Analysis of the distribution of argumentation-signalling items shows that all three English datasets broadly agree on the dominant metadiscursive function. The linguistic devices signalling inferential progression, argumentative consequence and legal deduction dominate all three datasets. This aligns with the previous studies discussed in the paper that CJEU judgements are strongly organised around an explicit transition from legal considerations to judicial conclusions. However, the MT systems do not completely replicate the patterns observed in the human translation because the baseline system relies more heavily on conclusion-oriented structures, while the fine-tuned model increases the proportion of contextual framing and qualification devices compared to human translations.

The statistical modelling of category distributions further demonstrated that the discourse organisation of CJEU judgements depends on a relatively small set of highly recurrent institutional templates. The overlap and token-mass analyses show that a limited number of sentence-initial and paragraph-initial expressions account for most of the argumentative organisation across English translations. At the same time, the systems diverged increasingly at lower-frequency levels. The overlap values decreased substantially once less frequent expressions entered the comparison, even though the shared high-frequency core continues to account for most of the token mass. Both MT systems demonstrate the same high level of overlap with the human dataset for the most frequent expressions (71%). Outside of the most frequent expressions, overlap decreases, with the fine-tuned system showing 46% overlap with human translations and the baseline system showing 34%. This suggests that both MT systems perform better when highly repetitive argumentative devices are used.

Finally, a case study based on the qualitative concordance analysis demonstrates that these differences are not merely lexical but reflect different translation strategies. Human translators often restructure French argumentative organisation reducing explicit inferential scaffolding, replacing subordinate framing structures with shorter connectors, or reorganising clause structure according to English institutional conventions. The MT systems, especially the baseline model, tends instead to preserve the formal inferential architecture of the French source texts much more

directly. This includes clause-initial hypotactic structures, complementiser-led discourse organisation, impersonal procedural framing and highly explicit inferential staging. The fine-tuned system is closer to human translations in this respect.

From the perspective of legal semiotics, these differences are significant because argumentative organisation forms part of the process through which legal meaning is constructed and communicated. Metadiscursive expressions do not simply connect textual units: they indicate how propositions are to be related and interpreted, for example as considerations, qualifications, justifications or conclusions. They therefore perform a pragmatic and rhetorical function by guiding the reader through the Court's reasoning and signalling the argumentative status of different parts of the judgement. The differences observed between human and machine translations can consequently be understood as differences in the semiotic organisation of legal discourse. The tendency of MT systems to preserve explicit inferential and hypotactic structures from the French source, while human translators more frequently compress or restructure them, results in different ways of presenting the relations through which the Court's reasoning is made interpretable. This does not necessarily alter the legal propositions expressed in the judgement, but it may alter the rhetorical and pragmatic organisation through which those propositions acquire their place and function within the judicial argument.

Returning to the original research question addressed by this paper, the findings provide empirical support for the manifestation of machine translationese at the level of argumentative metadiscourse. Firstly, unlike human translators, both MT systems reproduce the sentence and paragraph distributions of the French source texts almost exactly. Secondly, the MT systems preserve the inferential and hypotactic organisation characteristic of the French originals, particularly through clause-initial framing structures, complementiser-led constructions and explicit procedural scaffolding. Fine-tuning reduces some of these tendencies and partially shifts toward the patterns observed in the human English CJEU translations. Thirdly, while all systems share a highly recurrent institutional core of argumentative expressions, the MT outputs diverge from the human translations in the distribution and rhetorical realisation of lower-frequency discourse patterns. Because the human, baseline and fine-tuned translations are based on the same French source judgements, the systematic convergence of the two MT outputs in contrast to the human translations cannot be attributed simply to differences in the source texts. In terms of the

operational definition adopted in this paper, these recurrent machine–human contrasts therefore constitute evidence of machine translationese within the corpus examined.

These findings extend previous observations of machine translationese to the metadiscursive organisation of legal argumentation. Vanmassenhove et al. [4] associate machine translationese with the amplification of frequent patterns and the loss of less frequent alternatives, while Bizzoni et al. [3] show that machine translation exhibits translationese patterns that do not simply reproduce those found in human translation. The concentration observed here around a shared core of highly recurrent expressions, combined with divergence in the less frequent patterns, is compatible with these findings, although the present analysis concerns discourse-organising expressions rather than lexical or morphological complexity. The results are also consistent with Jiang and Niu's [6] finding that differences between human and machine translation extend to discourse coherence. The present study suggests that such differences can also be observed in the linguistic resources through which argumentative progression is signalled in institutional legal discourse.

The patterns associated with domain-specific fine-tuning in the present study should, however, be interpreted cautiously. The experimental design does not establish whether the greater formulaicity of the fine-tuned output and its partial shift toward patterns found in the human English translations result from the fine-tuning procedure itself, from characteristics of the domain-specific training corpus, or from the reinforcement of recurrent patterns characteristic of institutional CJEU discourse. These factors cannot be differentiated on the basis of the present comparison.

There are some clear limitations of the present study. Firstly, the present study focuses on only one legal genre, one language pair and one MT architecture. Additional research should expand upon all three of these factors in order to determine how widespread these tendencies are. Secondly, the present study is concerned with the investigation of general distributional properties of linguistic expressions. Only a small case study could be included alongside the quantitative analysis. For a more thorough understanding of how AI-based translation systems reshape the development of argumentation in legal discourse, large-scale qualitative studies with thorough annotation and the inclusion of human raters and potentially machine learning-supported annotation are necessary. This was beyond the scope of the present paper. To the best of my knowledge, this is the first study to compare a baseline and domain-specific fine-tuned NMT

system with institutional human translations in terms of the metadiscursive patterning of argumentation in EU case law. Although carrying out such studies is time-consuming and requires access to HPC, such studies are necessary for understanding the effects of AI-based translation systems on legal discourse. The relevant question is not only whether such systems translate legal texts accurately, but also whether and how they alter the discourse patterns through which legal reasoning is organised and communicated.

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